

# Homework 1

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## 1 Vowel space extraction and vocal-tract-length normalization

### 1.1 Formant Extraction

For each vowel token, we extract  $F_0$  and the first four formant frequencies ( $F_1$ – $F_4$ ) from a steady-state region of the vowel. We define the center of this region as the average of the two provided center-time annotations, i.e.  $t_c = \frac{c_1 + c_2}{2}$ , where  $c_1$  and  $c_2$  denote the two center-time annotations from the two judges. The steady-state window is set to 1/3 of the observed vowel duration,  $t_e - t_s$ , centered at  $t_c$ . Thus,

$$t_{\text{ss,start}} = \max\left(t_s, t_c - \frac{t_e - t_s}{6}\right), \quad t_{\text{ss,end}} = \min\left(t_e, t_c + \frac{t_e - t_s}{6}\right), \quad (1)$$

where  $t_s$  and  $t_e$  denote the observed vowel start and end, respectively.

Within this steady-state window, pitch and formant measurements are sampled with a hop size of 10 ms and averaged across all valid samples.  $F_1$ – $F_4$  are estimated using Parselmouth’s Burg formant tracker with a 25 ms analysis window and group-specific maximum formant frequencies of 5,000 Hz for men, 5,500 Hz for women, and 8,000 Hz for children (boy + girl).

Figure 1 shows four representative examples (one from each group). We plot the waveform, spectrogram with the steady-state window and extracted formant tracks overlaid. This provides a sanity check that the window is landing on the vowel and the tracker is following the right harmonics.

After extracting the formants, we compare them against the ground-truth values for  $F_1$  and  $F_2$ . We remove entries that have a ground-truth frequency of 0. Table 1 shows that the men group has the lowest MAE while achieving the highest correlation for both  $F_1$  and  $F_2$ , indicating higher accuracy in estimating the formants for men compared to children, who perform the worst.

Table 1: Formant extraction accuracy relative to the reference measurements. MAE is reported in Hz and correlation is Pearson’s  $r$ .

| Speaker group | $F_1$    |      | $F_2$    |      |
|---------------|----------|------|----------|------|
|               | MAE (Hz) | $r$  | MAE (Hz) | $r$  |
| Men           | 11.64    | 0.99 | 36.70    | 0.99 |
| Women         | 17.94    | 0.99 | 73.54    | 0.96 |
| Children      | 57.56    | 0.90 | 238.60   | 0.81 |

Figure 2 shows the scatter plots between the extracted and ground-truth  $F_i$  values, each with a  $y = x$  reference line and points colored by speaker group. From the scatter plots, both the men and women groups cluster closely around the reference line for  $F_1$  and  $F_2$ , while the children’s group is more scattered, with the extracted values tending to be higher than the ground-truth values.

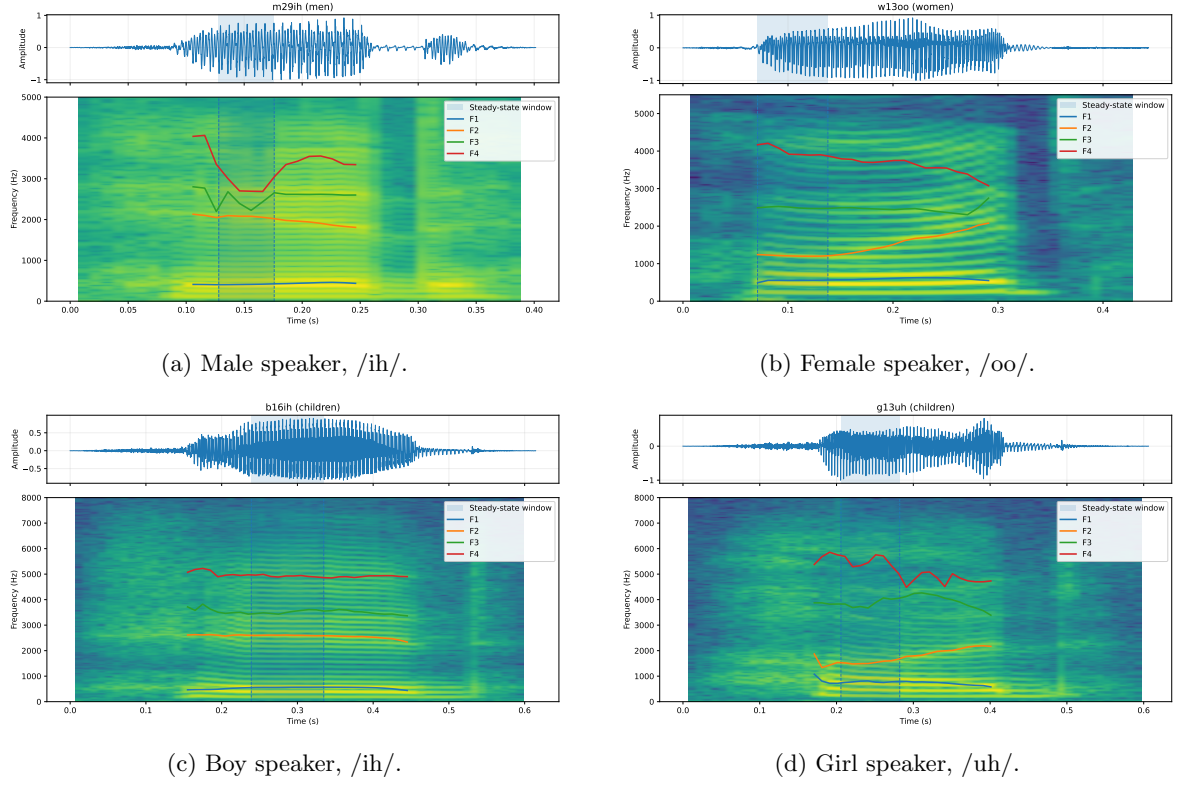


Figure 1: Example waveform and spectrogram visualizations with the steady-state window and extracted formant tracks. The examples span the three speaker groups: men, women, and children (boy + girl).

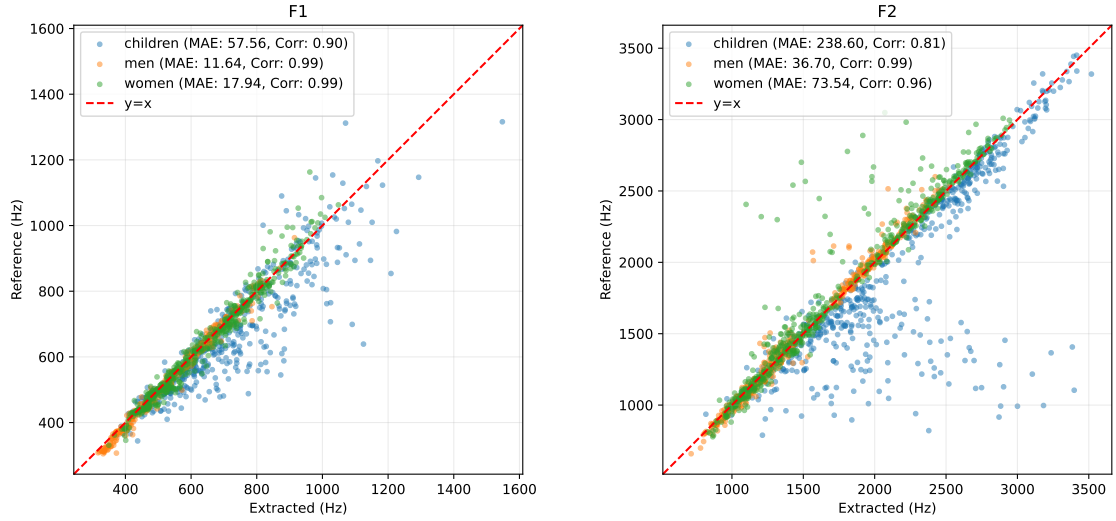


Figure 2: Extracted versus ground-truth  $F_1$  and  $F_2$  frequencies for each speaker group.

## 1.2 Group differences in vowel space

In this section, we plot the vowel space for each speaker group and compare the distributions of  $F_1$  and  $F_2$  across men, women, and children. Figure 3 shows the vowel-space distributions for the three groups, with each vowel represented by a different color. The axes are shared across all three plots to allow direct comparison between speaker groups.

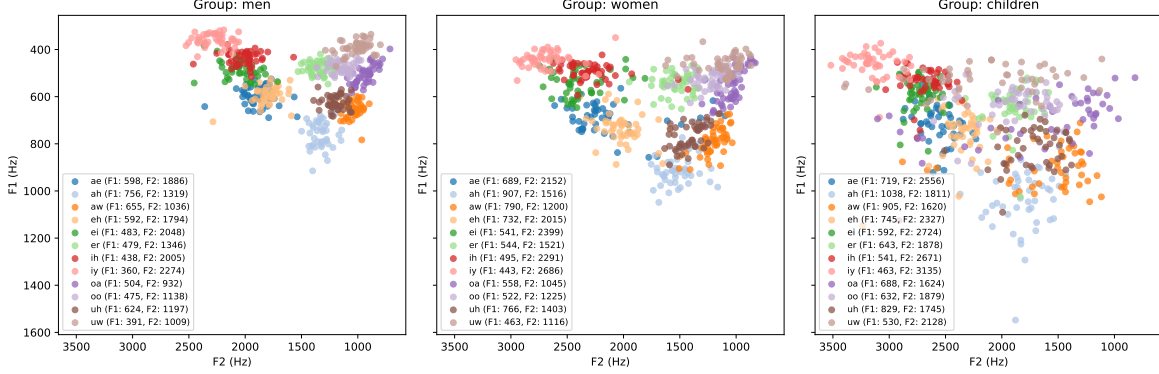


Figure 3: Vowel-space distributions for men, women, and children, with each vowel represented by a different color.

A clear shift in the vowel space is observed from the upper-right toward the lower-left when moving from men to women and then to children. For example, for /iy/, mean  $F_2$  increases from 2274 Hz in men to 2686 Hz in women and 3135 Hz in children, while for /ah/, mean  $F_1$  increases from 756 to 907 to 1038 Hz. This pattern reflects the generally lower formant frequencies of men and the progressively higher formant frequencies of women and children. The men's formant realizations are also more tightly clustered in the vowel space compare to children's realizations, which show a larger spread in absolute frequency (e.g. like /ah/ and /uh/).

## 1.3 Vocal-tract-length estimation

In this section, we perform vocal-tract-length estimation for each speaker using the formant-dispersion method. We first averages  $F_1$ – $F_4$  across all vowel tokens for each speaker and computes the mean spacing between consecutive formants, i.e.

$$\Delta F = \frac{(\bar{F}_2 - \bar{F}_1) + (\bar{F}_3 - \bar{F}_2) + (\bar{F}_4 - \bar{F}_3)}{3}.$$

The vocal tract length can be therefore estimated as:

$$L = \frac{c}{2 \times \Delta F}, \quad \text{where } c \approx 35000 \text{ cm/s}$$

Using this method and the extracted formants from 1.1, we observed the follow mean and std for each of the three speaker groups:

- Men:  $18.11 \pm 1.26$
- Women:  $15.51 \pm 0.81$
- Children:  $12.06 \pm 0.58$

, which aligns with the expected ranges (adult men 17–18 cm, adult women 14–15 cm, children shorter depending on age). Figure 4 shows the boxplot of VTL for each speaker group. We identify two outliers in the men's group that fall outside the IQR bounds (m44 and m40), as well as two outliers in the children's group (b18 and b07).

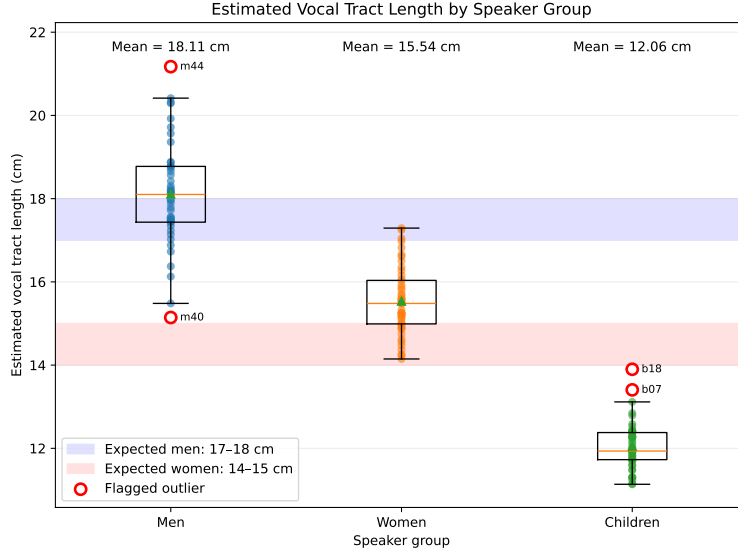


Figure 4: Estimated vocal tract length (VTL) distributions for men, women, and children. Shaded regions indicate the expected VTL ranges for adult men and women, and red circles denote speakers identified as IQR outliers.

#### 1.4 Formant normalization based on VTL

Using the vocal tract length calculated in the previous section, we perform vocal-tract-length normalization by multiplying each token’s  $F_1$  and  $F_2$  values by the ratio of that speaker’s VTL to the corpus-wide mean VTL. Specifically,

$$F_i^{norm} = F_i \times \frac{L_s}{\bar{L}},$$

where  $F_i$  denotes the original formant frequency,  $L_s$  is the VTL of speaker  $s$ , and  $\bar{L}$  is the mean VTL across all speakers.

Figure 5 shows the vowel space after normalization. Compared with Figure 3, vocal-tract-length normalization brings the  $F_1$  and  $F_2$  frequencies of the different speaker groups closer together, although an observable difference remains between the men/women and children groups.

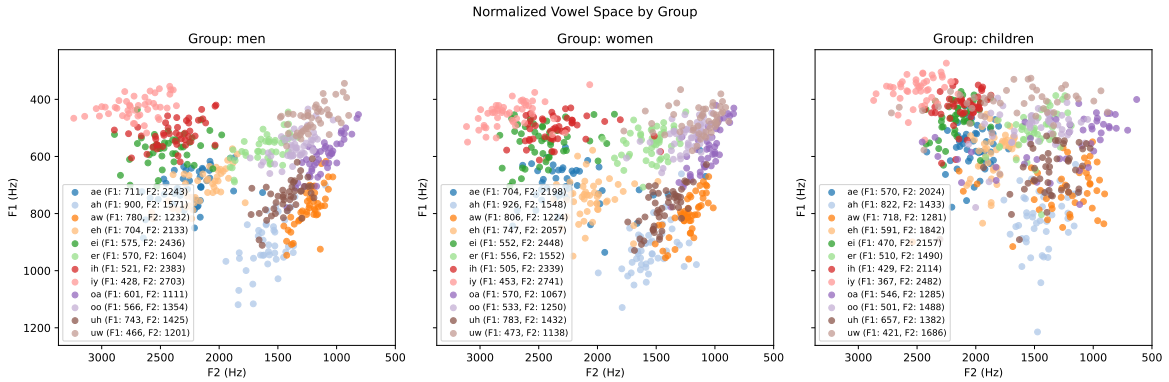


Figure 5: Vocal-tract-length normalized vowel-space distributions for men, women, and children.

To further quantify the distribution shift, we compute the pairwise centroid distance for each vowel

between each pair of speaker groups and compare the distances before and after VTL normalization. The percentage reduction in average centroid distance is defined as

$$\text{Reduction}(\%) = \frac{D_{\text{before}} - D_{\text{after}}}{D_{\text{before}}} \times 100, \quad (2)$$

where  $D_{\text{before}}$  and  $D_{\text{after}}$  denote the average pairwise centroid distances before and after VTL normalization, respectively.

From Figure 6, we can observe that the pairwise centroid distances are substantially reduced for all speaker-group pairs, with the men–women pair showing the largest reduction at approximately 78%. This indicates that VTL normalization brings the  $F_1$  and  $F_2$  distributions of different speaker groups closer together. However, there is still a noticeable gap for pairs involving the children group, suggesting that VTL differences only explain part of the  $F_1$  and  $F_2$  differences between adult and child speakers.

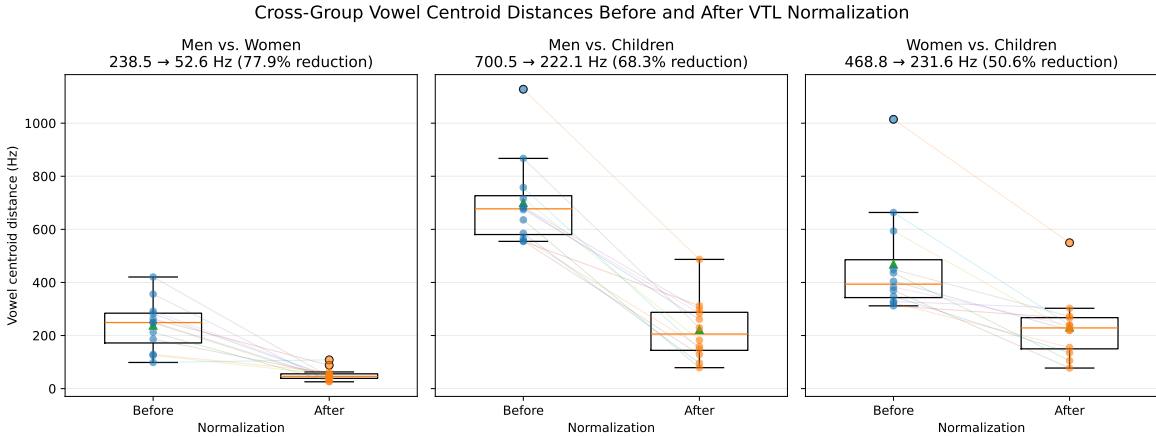


Figure 6: Pairwise vowel-centroid distances between speaker groups before and after VTL normalization. Each point represents the centroid distance for one vowel, and the lines connect the same vowel before and after normalization.

## 2 Vowel classification from acoustic features

### 2.1 Feature preparation

We assemble three feature tables:

- F1/F2: from your Sec. 1.1 extraction.
- F0/F1/F2: the same, with F0 added.
- MFCCs: 13-dim MFCCs, averaged across the same steady-state window used in Sec. 1.1.

### 2.2 Speaker-independent 5-fold cross-validation

To build the k-NN classifiers for predicting vowel identity using the three feature representations, we first select an optimal value of  $k$  using a held-out set to prevent data leakage. The held-out set consists of 20% of the men and women speakers. We choose  $k$  from the set  $\{1, 3, 5, 7, 9, 11\}$ . Since we use a single value of  $k$  across all three models, we select  $k = 3$ , which achieves the highest average held-out accuracy of 70.1% across the three feature representations.

We perform 5-fold cross-validation on the rest 80% of speakers. Figure 7 shows the confusion matrices summed across all five folds’ test predictions, along with the mean and standard deviation of the test accuracy across folds. The  $F_1/F_2$  features have the lowest average accuracy of  $0.643 \pm 0.037$ , followed by  $F_0/F_1/F_2$  with an accuracy of  $0.736 \pm 0.014$ . MFCCs achieve the highest average accuracy at  $0.737 \pm 0.014$ . For MFCCs, the most confused vowel pair is /ae/–/eh/, with 46 mutual confusions, followed by /ei/–/ih/ with 32.

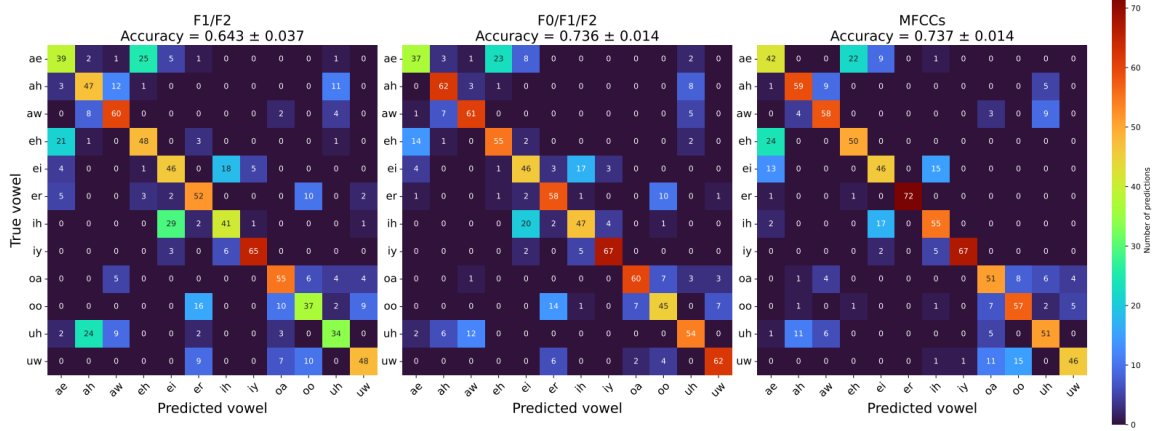


Figure 7: Confusion matrices for vowel classification using the three feature representations:  $F_1/F_2$ ,  $F_0/F_1/F_2$ , and MFCCs. Each matrix sums predictions across the five speaker-independent cross-validation folds.

We also plot the per-fold accuracies as box plots and perform a paired  $t$ -test between each pair of feature sets using the matched fold accuracies. Since performing multiple pairwise tests provides more opportunities to obtain a significant result by chance, using  $p < 0.05$  for all three comparisons inflates the false-positive rate. We therefore apply a Bonferroni correction to limit the probability of obtaining at least one false-positive result across the three pairwise comparisons. The corrected  $p$ -value is defined as

$$\hat{p} = \min(mp, 1), \quad (3)$$

where  $m$  is the total number of comparisons.

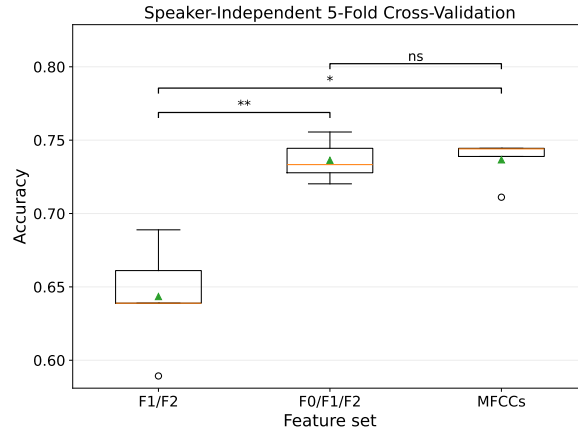


Figure 8: Speaker-independent 5-fold cross-validation accuracies for the three feature sets. Pairwise significance is tested using paired  $t$ -tests with Bonferroni correction.

Figure 8 shows significant differences between  $F_1/F_2$  and both  $F_0/F_1/F_2$  ( $\hat{p} = 0.0028$ ) and MFCCs ( $\hat{p} = 0.0162$ ). In contrast, there is no significant difference between  $F_0/F_1/F_2$  and MFCCs, with a Bonferroni-corrected  $p$ -value of 1.0.

### 2.3 Generalization to children: raw vs. VTL-normalized formants

In this section, we first evaluate the five models trained on the men and women data using raw  $F_1/F_2$  features on the children data, and compare their performance with five models trained using VTL-normalized  $F_1/F_2$  features. Figure 9 shows the boxplots of adult and child classification accuracies before and after VTL normalization. For raw  $F_1/F_2$ , the model achieves an accuracy of  $0.643 \pm 0.033$  on adults but only  $0.351 \pm 0.006$  on children ( $-0.292$  in accuracy). In contrast, using VTL-normalized  $F_1/F_2$ , the model achieves an adult accuracy of  $0.623 \pm 0.037$  and a higher child accuracy of  $0.416 \pm 0.004$  ( $-0.207$  in accuracy). However, a significant performance differences between adults and children speakers remains for both feature representations ( $p < 0.001$  for both raw and VTL-normalized  $F_1/F_2$ ).

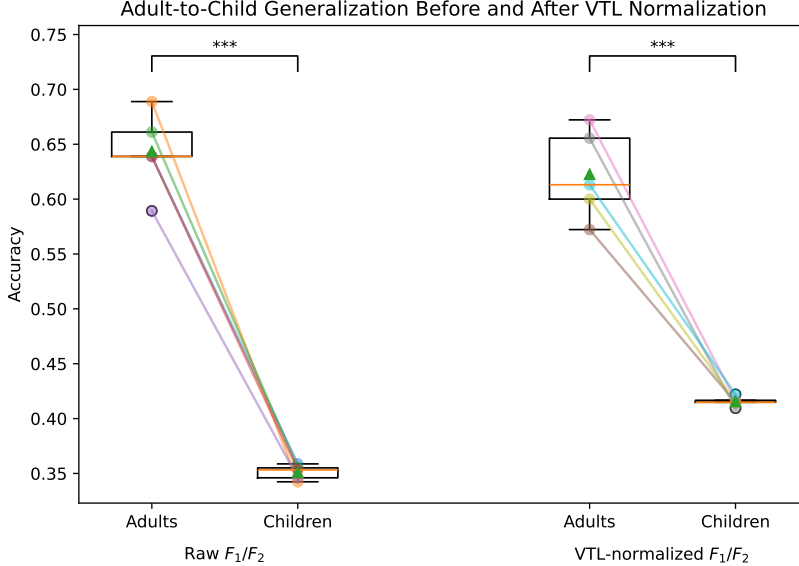


Figure 9: Adults and children classification accuracies using raw and VTL-normalized  $F_1/F_2$  features across five folds.

To determine whether VTL normalization significantly reduces the adult-to-child performance gap and improves generalization to children, we perform a paired  $t$ -test on the fold-wise decreases in accuracy from adults to children before and after normalization, i.e. comparing  $(Acc_{\text{adult}}^{\text{raw}} - Acc_{\text{child}}^{\text{raw}})$  against  $(Acc_{\text{adult}}^{\text{VTL}} - Acc_{\text{child}}^{\text{VTL}})$  across the five folds. We obtain a  $p$ -value of 0.0028, indicating that the reduction in the adult-to-child performance gap differs significantly between models trained on raw  $F_1/F_2$  features and those trained on VTL-normalized  $F_1/F_2$  features. Since the average performance drop is smaller after VTL normalization, VTL-normalized  $F_1/F_2$  significantly reduces the adult-to-child generalization gap, although an observable performance drop on children still remains.

### 2.4 MFCCs and generalization

Lastly, we repeat the analysis from the previous section using MFCCs as the feature representation, as shown in Figure 10. MFCCs demonstrate the best generalization to children, with accuracy decreasing

from  $0.737 \pm 0.014$  on adults to  $0.631 \pm 0.010$  on children ( $-0.106$  in accuracy), although a significant difference still exists between adults and children performance.

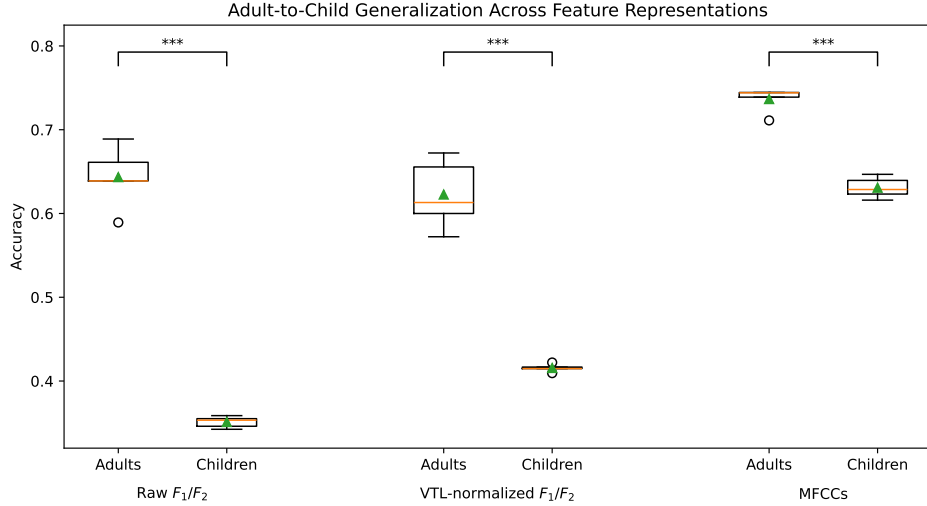


Figure 10: Adults and children classification accuracies across different feature representations.

To compare the performance drop of MFCCs against that of raw  $F_1/F_2$ , we conduct a paired  $t$ -test and obtain a  $p$ -value of 0.0007. This indicates that the difference in performance drop is statistically significant, with MFCCs showing a smaller adult-to-child performance drop and therefore better generalization to children.

One reason why MFCCs still perform better than  $F_1/F_2$  even after VTL normalization is that MFCCs encode the overall spectral envelope, while  $F_1/F_2$  use only two scalar values. This low-dimensional representation may not capture enough information to distinguish vowels robustly across speaker groups, even after normalizing for vocal-tract-length differences. Figure 5 also shows that noticeable differences in the  $F_1/F_2$  distributions between children and adult speakers remain after VTL normalization.

When comparing the performance between using  $F_0/F_1/F_2$  and  $F_1/F_2$ ,  $F_0/F_1/F_2$  achieves higher accuracy on adults ( $0.736 \pm 0.01$ ) and also higher accuracy on children during generalization ( $0.399 \pm 0.01$ ). However, the adult-to-child performance drop is larger for  $F_0/F_1/F_2$  (0.338) than for  $F_1/F_2$  (0.292), indicating that although adding  $F_0$  improves in-domain classification accuracy, it does not improve robustness to the adult-to-child domain shift.

## Code

The code used to reproduce the experiments, results, and figures in this report is available at:

[github.com/Morris88826/CSCE630\\_Speech\\_Processing/tree/main/hw1](https://github.com/Morris88826/CSCE630_Speech_Processing/tree/main/hw1)